Coherent Cross-View Syndromes for Credit-Card Fraud: Mechanism Validation Under Classical Stop Gates

2026 Global Quantum + AI Challenge – Phase 1 Concept Proposal

HSBC: *Quantum-Enhanced Credit Card Fraud Detection for Digital Payment Ecosystems*

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https://github.com/hanoipho997/hsbc-phase1-r3

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1. Decision problem and proposal thesis

Fraud detection needs a probability-like score, a binary decision, per-transaction attribution, and unchanged-prevalence evidence. We use AUPRC as the primary ranking metric because it is recommended for extreme imbalance, alongside AUC-ROC, precision, recall, F_1, calibration, and recall at fixed review budget.

We propose a compact, circuit-realizable coherent anomaly and syndrome mechanism, then subject it to classical and QML twins. The fixed-L=1 mixer increment replicates, and angle adaptation beats a prespecified random-freeze distribution. The scorer is not statistically resolved from our strongest product quantum kernel or hybrid QNN, while competent classical models remain as good or better. **We claim no quantum advantage.** Phase 2 is a capped Amazon Braket mechanism-validation study with exact classical twins and stop gates, not a promise to improve a production fraud backbone.

Three claims, in order of strength.

1. **Verified mechanism.** In a fresh-ancilla binary stacked-LCU implementation, the joint pass/fail pattern is an interpretable coherent measurement. Complete path dephasing at the declared insertion point makes its pattern law input independent. The endpoint, sector algebra, and an X-readout success-derivative identity have exact and circuit referees.
2. **Bounded named-instance competitiveness.** On the frozen ULB split, stacked-LCU's differences from product-fidelity QSVC and a Deloitte/AWS-shaped hybrid QNN are unresolved; equivalence is not established, and classical point estimates are as good or better. This is a common-split predictive-quality bracket, not a matched-runtime or matched-hardware-resource tournament.
3. **Application result.** On the prospectively frozen IEEE-CIS chronological experiment, every gate failed and the quantum residual harmed AUPRC. The classical backbone remains the detector; the circuit is monitoring-only unless a future no-harm and benefit gate passes.

Prior fraud-QML metrics use materially different, often balanced or reduced protocols. We do not infer how external models would perform under ours. The deployed-size circuit and its exact 2^L path twin are classically tractable; no result below shows runtime, energy, sample-complexity, or FTQC advantage.

2. Technical approach

Architecture and coordinates. Twelve semantic-view features are quantile-angle encoded on 12 qubits. Three binary LCU layers use one fresh ancilla each. Layer j implements

$$K_j = \cos^2\beta_j U^{(V)} + \sin^2\beta_j e^{i\phi_j} U^{(W)}, \quad F_j = \cos\beta_j \sin\beta_j \left(e^{i\phi_j} U^{(W)} - U^{(V)}\right), \tag{1}$$

where $U^{(V)} = e^{-i\theta H_V}\,\mathrm{RX}_V(\varphi)$. Hence $\partial_{\beta_j} K_j = 2F_j$. This is an exact X-readout derivative identity, not a new general differentiation method. Engine A uses softmax logits (ℓ_0, ℓ_1) with

$$\partial_{\ell_0} K = -\sqrt{a_0 a_1} F, \qquad \partial_{\ell_1} K = +\sqrt{a_0 a_1} F. \tag{2}$$

The common-logit direction is a gauge: 18 raw parameters, 15 identifiable. Khatri–Zohren–Matos already define S-LCU layering and study its trainability and simulation scaling [5]. We claim neither architecture nor name; our setting-specific object is the fresh-ancilla measurement package.

Syndrome and twins. One ancilla Z setting yields eight bins (seven independent values) for checks (A,B), (B,C), (A,C); finite shots are required. Complete dephasing gives the input-independent product of Bernoulli($2a_0a_1$) under fresh ancillas, unitary branches, exact dephasing after SELECT/before PREP†, and no later ancilla operation. The exact 2^L path expansion is the mandatory classical twin.

Training and explanation. Mixing-angle gradients use one X setting per layer; branch parameters use finite-Fourier shifts. The measured-flag Fisher metric uses $1 + L = 4$ settings for the deployed $L = 3$ mixing-angle/flag block versus 13 for the full conditional-state construction. This is not a uniform shot advantage or all-parameter count. Explanations combine legit-fitted view Ising tables, named syndrome failures, local sensitivity, and classical-backbone SHAP.

3. Evidence, strongest first

3.1. Verified mechanism and Braket-local bridge

Dense algebra and circuit referees reproduce states, sectors and derivatives at 10^{-13}–10^{-16}. The dephased endpoint agrees with its formula to 8.3×10^{-16}. A Braket-SDK **local** density-matrix run of a 6-system+3-ancilla instance agrees with the exact twin to 3.3×10^{-15}; this is not SV1, DM1, a managed simulator, or QPU result.

The original interval code deduplicated bootstrap rows and did not redraw shots. The transparent post-outcome correction retains duplicates, nests 2,000-shot multinomial draws, and uses 99.5% intervals for the intended ten-decision family:

seed	900	901	902	903	904
exact $W(0) - W(1)$	.07396	.03174	.05308	.05527	.04489
99.5% lower bound	.02062	.01318	.01583	.01485	.01413
99.5% upper bound	.11659	.04614	.08164	.08728	.06890

Only a seed-900 plug-in noise sensitivity exists: low-noise interval $[.00596, .02212]$; high-noise $[-.00023, .00049]$. The intended five-seed device-noise gate is unresolved. This motivates, but does not complete, a hardware experiment.

3.2. ULB: ablation, OCC parity, and named rivals

S1 is the analytic one-layer LCU/imaginary-time-inspired anomaly score without a trained mixer; S2-L1 is its fixed-depth trained-mixer form; OCC fits only legitimate training rows. On the unchanged-prevalence holdout (56,962 rows, 99 frauds), AUPRC progresses 0.264 hard bits, 0.688 soft quantile encoding, 0.696 S1, and 0.704–0.706 S2-L1. The dominant gain is classical encoding.

The fixed-L=1 mixer increment replicates at $+.007242$ $[+.001366, +.013372]$. Trained minus frozen-random is $+.120265$ $[+.084777, +.153449]$, while frozen-random minus matched $\phi = 0$ is $-.106891$ $[-.137744, -.072477]$. All ten trained-minus-random point signs are positive; intervals condition on the ten fitted seeds. This is a trained fixed-protocol effect, not a generic coherence, depth, or advantage claim.

PCA OCC reaches 0.7238 and IsolationForest 0.6967, matching or beating the standalone scorer. The bounded QML bracket shares split, features, and test rows with family-specific caps, but no common runtime/shot/query currency. Stack reference S2-L1 is 0.705758; simultaneous intervals cover seven primary comparisons:

named implementation	AUPRC	stack minus arm	classical/ablation twin
product-fidelity QSVC	.701587	$+.004171$ $[-.050248, +.068656]$	RBF-SVC .705807
Deloitte/AWS hybrid QNN	.707566	$-.001807$ $[-.035944, +.033507]$	dense head .721783
ring-IQP Nystrom	.542580	$+.163179$ $[+.048194, +.270152]$	RBF-Nystrom .730616
projected ring-IQP SVC	.632009	$+.073749$ $[-.026860, +.169265]$	RBF-SVC .705807

Product-QSVC and hybrid-QNN differences are unresolved; neither meets $\pm.01$ equivalence. Local 80%-power MDEs are .0748 and .0435 AUPRC, conditional on these paired score distributions. Reduced-effort VQC/QAE/AE stress tests remain implementation-qualified in the audit, not family proof.

Supervised-screened dominant-mode S1/XGB AUPRC is .944/.228 ($k = 3$), .945/.413 ($k = 4$), and .945/.394 ($k = 6$), with paired differences $+.7162$ $[+.5973, +.8122]$, $+.5319$ $[+.3829, +.6736]$, and $+.5507$ $[+.4152, +.6863]$. Fraud labels determine the screen, centroids, and dominant mode; minority modes mostly reverse and a legitimate-only screen collapses. This is off-manifold sensitivity, not label-free novel-fraud advantage. At 128 shots, mean AUPRC delta is $-.00483$, but the 30-draw range $[-.04596, +.03528]$ leaves direction unresolved.

3.3. IEEE-CIS Engine A: prominent FAIL

The core split, routing, endpoints, gates and outcome wordings were frozen before sealed-test access; feature/view and arm design used train/validation data. A 144-fit XGBoost+LightGBM backbone reached validation AUPRC .65195, then .52939 on the chronological sealed period.

ENGINE-A VERDICT: FAIL G1–G4. Quantum minus backbone AUPRC is $-.002298$ $[-.004087, -.000682]$. Quantum minus the strongest matched classical residual is $-.001265$ $[-.003144, +.000476]$, unresolved and not equivalent. Keyed noise is $-.007168$ $[-.010463, -.004060]$. No residual is deployed.

At $K = 400$, frozen validation-band oracle headroom was only .8675 recall percentage points. A secondary rolling-origin analysis is consistent with cross-region score interleaving, while both locally frozen in-period splice contrasts were unresolved. Noise excludes a quantum-specific explanation but does not identify finite-sample overfit, reranking, calibration, or drift as the cause.

3.4. Conditions map

axis / threshold	point winner	evidence and boundary
five total fraud labels	classical OCC	IF .587; XGB-full .546; S1 .483; wide seed ranges
later chronological ULB 20%	classical supervised	XGB-full .792; XGB-8 .786; S2 .753
screened dominant mode k = 3/4/6	S1, classically exact	.944/.945/.945; minority modes reverse
prevalence .02%, reweighted	S1, classically exact	S1 .498; XGB-8 .488; IF .336; no row subsampling
inward shift $\epsilon = .05/.10/.20$	S1, classically exact	.592/.580/.450 vs XGB .549/.365/.358; not attack feasibility
Sparkov supplied test	classical supervised	XGB .8721; IF .0640; S1 .0062
ULB↔Sparkov	no useful transfer	best weak OCC .0537; coordinates do not transfer
IEEE temporal holdout	backbone / no residual	backbone .52939; Q residual −.00230

Verdict: classical coverage suffices everywhere measured. Rows that favor S1 are exactly classically evaluable at tested depth. No measured condition requires a quantum processor.

4. Outputs, explainability, and impact

The deployable output is the classical backbone probability in $[0, 1]$ plus a validation-chosen binary decision. On visible IEEE-CIS validation data, Brier is .02135 versus .03752 for constant prevalence and ten-bin ECE is .01270; no post-hoc recalibration was fitted. At frozen threshold .994177, TP/FP/FN/TN is 223/14/4388/113483 (precision .9409, recall .0484, F_1 .0920). Derived only from existing sealed aggregates, without a new test read, it is 167/0/3897/114044 (precision 1.000, recall .0411, F_1 .0789).

An artifacted, label-blind validation illustration uses row token `cb7b9f51cf78481f`: backbone probability .9071, in the review band. Mean failure probabilities are .347 for A–B, .136 for B–C, and .547 for A–C; P(any failure) = .781, with A–C-only the most likely nonzero pattern (.347). This deliberately extreme model-selected case is not a representative-case or accuracy claim.

Expected Phase-2 impact is bounded: a Braket mechanism demonstration; a no-harm monitoring/attribution rail; a public full-prevalence evaluation template; and conditions that tell HSBC when to stop. It is not a promise of fraud-score improvement.

5. Phase-2 Braket work package and stop gates

1. **Dephasing witness.** Materialize $\mathcal{D}_q$ as randomized physical I/Z blocks, assigning Z with probability $q/2$; simulator `phase_flip` cannot run on a QPU. Freeze device, native compile, topology, calibration, job order, data approval, power and multiplicity. The reduced base is 80,000 shots before twirl, identical-branch and calibration controls.
2. **X-readout estimator.** Compare against optimally allocated parameter shift, SPSA and a Hadamard test at matched state preparations and shots; report cosine, bias, variance and shots-to-target.
3. **Flag geometry.** Replicate flag-NG vs Adam under calibrated noise, counting shots rather than settings. Existing $+.001882$ $[+.000239, +.004057]$ is a small simulator optimizer effect.
4. **Native-resource gate.** Compile for the selected Braket device's actual gates/connectivity, then power and price the complete experiment.
5. **Application gate closed.** Reopening needs a new externally timestamped protocol, temporal rehearsal, rank-preserving integration, calibration, and no-harm gate. The IEEE sealed period gets no further confirmatory read.

Stop if references fail, the contrast is underpowered, calibrated noise erases it, or the classical twin matches the operational benefit.

6. Deployment, governance, and team

The backbone ranking is never overwritten. Syndrome patterns are an asynchronous monitoring/attribution rail; exact classical paths are the fallback. IEEE-derived row arrays are `PRIVATE_LICENSED` and excluded from public release, VM handoff, or AWS transfer absent licence and governance approval. Public artifacts contain aggregate JSON, theorem code and synthetic referees.

Ha Cong Nguyen is the single-member team, lead/contact, model developer and model-risk owner, responsible for evidence integrity, classical stop gates, calibration/drift monitoring, and requiring independent validation before any production influence. The work demonstrates adversarial self-audit, OCC parity, sealed-test discipline, unchanged-prevalence rivals, circuit referees, negative-result publication, and a versioned post-outcome correction. Affiliation, short biography and publications remain submission-form metadata if required. The sanitized release record is `https://github.com/hanoipho997/hsbc-phase1-r3`.
Scientific-content commit:
`0bb260f21814a2d2aed9c1b3b5cf53b0d6e87cf0`.
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Appendix A – Evidence and scope ledger

Parameters and measurements.

instance	raw / identifiable	measurement/result boundary
ULB $L = 1$	4 / 3	source/new-seed AUPRC about .704/.706
original ULB $L = 2$	8 / 6	selected result .648; depth did not help
Engine-A cross-view $L = 3$	18 / 15	eight bins/seven independent values, one Z setting

Implementation-qualified stress tests. Our re-uploading VQC (.00648), one-layer trash-occupation surrogate (.00103), kernel OCC (.00451), and small hybrid latent AE (.06435) performed poorly but do not bound their families: architecture/tuning, polarity, native-FPR, or capacity controls were insufficient. They are audit stress tests, not headline competitors.

Theory floor. (i) For the tested commuting core/product encodings, classical estimators match or improve the reported shot variance. (ii) For an unamplified diagonal filter, $p_{\mathrm{s}}P(z^*|\text{success}) = q_0(z^*)g(E(z^*))$ is a raw-hit-yield identity, not a universal no-go. (iii) The LCHS period/span rule is a necessary alias-free precheck: failure permits but need not force inversion, and passing guarantees neither monotonicity nor accuracy. (iv) The dephasing theorem is scoped to the assumptions in Sec. 2; its curve is classically exact here. (v) Flag-NG geometry is known; matched-shot noisy behavior remains open.

Generic compilation envelope. Qiskit O3, ten seeds, exact 15-qubit operator: 155 RXX on abstract all-to-all RXX; 238 CZ on abstract all-to-all CZ; 354–374 CZ on a synthetic full 4×5 square grid; 486–513 CZ on a line. These are not vendor-native or equal-fidelity results; the grid is not an IQM Garnet topology. Sze et al. provide trapped-ion multiplexer precedent, not validation of these counts [10].

FTQC boundary. Coherent polynomial-optimization and linear-solve modules support a separate roadmap, not Phase-1 advantage or parameter-space QNG. Any claim needs end-to-end oracle, conditioning, precision, success, non-tomographic readout, and best-classical accounting.

References

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Appendix B – Reproducibility and release status

The Engine-A sealed result is cited from `runs/hsbc_challenge/engine_a_v1/unblind_real_v1.json`; "one read" applies only to that IEEE-CIS sealed partition. ULB test rows were reused across declared audits. Revision-3 additions form a fail-closed sanitized release package. Its scientific-content commit is frozen before the final metadata-only commit and Zenodo archive. Output completeness is frozen in `docs/hsbc_challenge_revision3_output_protocol_v1.md`, generated by `scripts/hsbc_challenge/report_revision3_outputs_v1.py`, with aggregate artifact under `runs/hsbc_challenge/revision3_v1/`. No sealed row or score was reread; IEEE-derived row arrays remain private. Engine-A implementation/evidence is source-locked at commit `5242b2d`; the Revision-2 base is `2fe355f`; sanitized Revision-3 scientific-content commit:
`0bb260f21814a2d2aed9c1b3b5cf53b0d6e87cf0`.

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